ACTIVITY 1.7 Measurement and units

SKILLS

- Numeracy
- Processing and analysing data and information
- Scale and measurement

1 Complete the passage below by using the words in the list.

distance other paper ruler thermometer body mass duration measured heat amount

Length measures the amount of _____ between two points. The length of an A4 sheet of _____ is about 29.7 cm or 0.297 m. It is usually measured with a _____ or tape measure.

Time measures the _____ of an event or between two events. The time taken for a person to walk from one side of a room to the _____ is about 5 s. It is usually _____ with a watch.

Mass measures the _____ of matter in an object; it is sometimes called weight, but not in Science. The _____ of an apple is about 100 g or 0.1 kg. It is often measured with a balance.

Temperature measures the amount of _____ energy in an object. The temperature inside your _____ is about 37.5 °C. It is usually measured with a _____.

2 Complete the table below.

Description of what is to be measured	Instrument used	Measurement (with unit)
Height of a beaker		
Time for 100 heart beats		
Temperature of tap water		
Volume of a steel bolt		
Mass of a light bulb		

3 Suggest ways to calculate the following.

a The average mass of a typical pebble, using four pebbles.

_____

_____

b The average time taken for a feather to reach the ground when dropped from a table top, using four trials.

_____

_____

4 For each of the following, determine what is being measured (e.g. mass, length, time, temperature), and then calculate the average.

a 6 cm, 7 cm, 12 cm

_____

b 10 g, 15 g, 16 g, 17 g

_____

c 25 °C, 27 °C, 31 °C, 32 °C

_____

d 60 s, 55 s, 35 s, 70 s

_____

5 What is the standard unit for measuring:

a mass? _____

b time? _____

c temperature? _____

6 Why are different instruments used for measuring mass, such as bathroom scales, a spring balance and an electronic balance?

_____

_____

7 Complete the missing spaces for converting measurements. The first one has been done for you.

1 kilogram	1000 grams	1 000 000 milligrams
4 kilograms	grams	4 000 000 milligrams
kilograms	grams	3000 milligrams
kilograms	grams	9000 milligrams
kilograms	7 grams	milligrams
kilograms	grams	1600 milligrams
0.01 metres	centimetres	10 millimetres
metres	3 centimetres	millimetres
metres	centimetres	140 millimetres
2.3 metres	centimetres	millimetres